1.2 Assumptions, data collection and generation

The analysis gives a probability of failure P_f and a reliability index β for each walmuur (the low wall between the werf and the canal) that still stands on its original timber pile foundation. This needs, per wall: whether the foundation has been renewed, the geometry of wall and foundation, the retained height, the soil and water behind the wall, the load on the werf and the condition of the timber piles. The municipal asset register lists the walls, their length and age, but none of the structural data. We therefore collected open data and project documents, derived the missing values with scripts, and filled the remaining gaps with stated assumptions. Within the time of the project, no data were requested from the municipality.

1.2.1 Data collected

Table 1 lists the data, what each source provides and its main weakness.

Table 1: Data used for the walmuren, with source and main limitation.

Data need	Source	Used for	Limitation
Walls, length, age	Asset register [5]	51 walmuren; screening	No height or foundation data
Werf level	AHN 0.5 m [14]	Retained height H	Canal bed not measured
Soil	71 DINO borings [16]	Soil class, sets K and γ	20–144 m away; often disagree
Groundwater	BRO wells [2]	GHG and GLG	Nearest well 143–418 m
Traffic	BGT, NDW [11, 12]	Road distance, axle limit	Road is not on the werf
Foundation	Programme reports [1, 6]	Renewed or timber	8 walls unclear
Wall type	TNO dive notes [15]	P(type 2) per rak	Not known per wall
Section, piles	[4, 3, 9]	Shapes, levels, piles	Schematic; Oudegracht
Timber, method	Hemel [7, 8]	Pile strength, decay, load	Amsterdam piles
Target β	NEN 8700 [13]	1.8 / 2.5 / 3.3	Class not chosen

The foundation data decide which walls are modelled at all. Of the 51 walmuren, 32 are already renewed on steel, including 11 of the 29 that the register marks as past their design life. Of the 19 on timber, 2 have no werf: the street runs to the water, which is a different structure. This leaves 17 walls (611 m) for the P_f calculation (Appendix A).

1.2.2 Data generation

Eight scripts turn the raw data into one input row per wall. They run in a fixed order, each reading the output of the one before (Appendix A). The main generated values are:

- **Werf level and H:** the median of six AHN points on the werf side of the wall (+0.99 to +1.57 m NAP). H is the werf level minus the wall underside at −0.08 m NAP [3]; median 1.22 m.
- **Soil class:** the share of clay-like and sand-like soil between werf and wall underside in the nearest boring within 250 m. At least 70 % of one kind gives that class, otherwise “mixed”. Of the 51 walls, 23 are clay/peat, 16 sand and 12 mixed.
- **Foundation status:** a hand-compiled table of renewals per rak and bank, linked to the register through the city’s *Werfmuren* map.
- **Wall type:** divers found a timber beam (*watersloof*, only in type 1) or a concrete floor (only in type 2) in some rakken [15], which gives a chance of type 2 per rak.
- **Monte Carlo samples:** 400,000 draws per wall and wall type, with a fixed seed, for 2026 and 2041 (end of the 15-year reference period of NEN 8700).

Every raw file is listed with its source and download date, so all results can be regenerated from the raw data.

1.2.3 Assumptions

Where no data exist we use the assumptions in Table 2. Uncertain values get a distribution, not a single value. All 27 inputs are listed in Appendix A.

Table 2: Main assumptions where data are missing.

Quantity	Assumption	Basis
Canal bed	At the wall underside (-0.08 m NAP), so H is a lower bound	[3]
Wall type	P(type 2): 0.2 (rak 3–6), 0.5 (rak 21), 0.9 (rak 22), 0.8 (rest)	[15, 4]
Cross-section	Two measured drawings for all walls; floor level ± 0.15 m	[4, 9]
Pile condition	Soft shell $U(0, 45)$ mm, all walls; grows 0.10–0.25 mm/yr	[7]
Piles	Diameter $N(180\,\text{mm}, 0.10)$; spacing $U(0.8, 1.2)$ m	[9, 7]
Soil	K lognormal and γ normal per class, independent	Group report, Tab. C.5
Water	Canal $+0.6$ m NAP; groundwater $U(+0.5, +0.9)$ m NAP	[3]
Surcharge	$q \sim N(7\,\text{kPa}, 0.20)$ at a distance $U(0, 3)$ m	[8]
Proven strength	Keep only the draws in which the wall stands today	[8]

1.2.4 Effect on the results and validation

The assumptions do not affect the results equally. In order of effect on β (interpreted further in Section 4.2):

1. **Wall type.** For the same wall, type 1 gives a β that is 1.4–2.4 lower than type 2. With the chance per rak, β is 0.1–1.5, so all 17 walls are below 1.8. If all were type 2, 7 of the 17 would pass 1.8. Whether a wall passes therefore depends mostly on its type, which the original drawings or a dive inspection can settle.
2. **Pile condition.** For type 2 the piles fail far more often than the wall overturns (median P_f 0.075 against 0.004). The same broad shell is used for every wall, so it shifts all P_f but not the ranking.
3. **Retained height and soil.** The soil moment grows with H^3: 0.1 m more height adds about 25 %. As H is a lower bound, this assumption makes P_f too low. Clay/peat gives 43 % more soil load than sand, and nearby borings agree for only 9 of the 51 walls.
4. **Surcharge.** 7 kPa is heavy for a pedestrian werf, so this assumption makes P_f too high.

We checked the data and assumptions where possible. The median AHN werf level on the Oudegracht ($+1.13$ m NAP) matches the $+0.95$ to $+1.15$ m NAP measured in [4, 3]. The renewed length in our foundation table (about 3.36 km) matches the city's 3,356 m. The strongest check is that all 17 walls stand today. A first model that treated the wall as a masonry block gave $P_f \approx 1$ and was rejected. With the measured sections, a type-2 wall stands in 87–100 % of the draws, a type-1 wall in only 10–98 %. Either the type-1 model is conservative or many of these walls are type 2; the proven-strength update accounts for this.

Settlement from pile rot, the damage actually seen on these walls [4], and failure of the timber screen are not modelled. The β values should therefore be read as an indicative ranking and order of magnitude, not as exact values. The data that would improve them most are the wall type per wall, the measured pile condition and canal soundings.

2.1 Probabilistic models

The risk analysis (Sections 3.1 and 3.2) needs the probability of failure of each of the 17 walmuren that still stand on timber piles (Section 1.2). We model it with two limit states, evaluated by Monte Carlo simulation, and check the model against data and observations. The full equations are in Appendix B.

2.1.1 Limit states

A wall fails when its body (wall, floor and the soil on it) overturns about the front pile row, or when a timber pile breaks in bending [7, 8]:

$$g_1 = \theta_R M_R + M_\delta - S, \tag{1}$$

$$g_2 = \mathrm{MOR} - \left(\theta_M \frac{32M}{\pi d^3} + \frac{4N}{\pi d^2}\right), \qquad d = D_0 - 2(s_0 + r\,(t - 2026)). \tag{2}$$

S is the moment of soil, water and surcharge, M_R that of the weight of the body and M_δ that of wall friction. M and N act in a pile whose sound core d shrinks as the rotted shell s_0 grows at rate r. The wall fails if either limit state is violated (series system): $P_f = P(g_1 < 0 \cup g_2 < 0)$ and $\beta = -\Phi^{-1}(P_f)$, estimated as the fraction of failed draws (400,000 per wall and type; 2026 and 2041).

2.1.2 Random variables and distributions

Table 3 lists the random variables. Each type of distribution follows from what is known: a *lognormal* for positive quantities with a multiplicative error, such as K and the model factors θ [10]; a *normal* for unit weights and timber properties, where a CoV of at most 0.20 makes negative values negligible; a *uniform* where only a range is known, as it adds nothing beyond the bounds; and a *Bernoulli* for the wall type.

Table 3: Random variables of the model. Per wall: values differ between walls. CoV = coefficient of variation.

Variable	Distribution	Parameters	Estimation
Werf level (per wall)	Normal	AHN median; $\sigma = 0.05$ m	AHN data
Wall type T (per wall)	Bernoulli	$P(T{=}2) = 0.2$–0.9 per rak	Dive notes [15]
Earth pressure K	Lognormal	0.35–0.60; CoV 0.20–0.25	Moments per soil class
Soil unit weight γ	Normal	15–18 kN/m^3; CoV 0.05–0.06	Moments per soil class
Surcharge q; distance a	Normal; uniform	7 kPa, CoV 0.20; 0–3 m	[8]; range
Groundwater	Uniform	+0.5 to +0.9 m NAP	Range around canal level
Unit weights wall	Normal	19 / 23 kN/m^3; CoV 0.05	Typical values
Floor level Δz	Uniform	± 0.15 m	Drawing accuracy
Model factors θ_R, θ_M	Lognormal	mean 1; CoV 0.15, 0.10	Judgement; JCSS [10]
Pile diameter D_0	Normal	180 mm; CoV 0.10	[9, 7]
Pile spacing s	Uniform	0.8–1.2 m	[9]
Soft shell s_0; growth r	Uniform	0–45 mm; 0.10–0.25 mm/yr	Measured range [7]
MOR; E	Normal	22 MPa, CoV 0.10; 9 GPa, 0.20	[7]
Subgrade modulus k	Uniform	4,300–10,000 kN/m^3	[7]

The parameters are estimated in three ways: from a literature mean and CoV (method of moments), from our own data (werf level, wall-type prior), and by Bayesian updating with the observation that every wall still stands (proven strength [8]). Draws in which the wall would already fail today under soil and water alone are rejected; the same evidence updates $P(T{=}2)$ (Appendix B).

2.1.3 Dependence between the variables

The dependence is modelled explicitly in four ways:

1. **Common cause.** The wall type sets the geometry, the floor level, the rake of the piles and the head fixity together. The soil class sets the means of K and γ. P_f is a mixture: $P_f = \sum_T P(T \mid E)\, P_f(T \mid E)$.
2. **Shared variables.** One draw of K enters the soil pressure, the surcharge pressure and the wall friction ($\delta = \frac{2}{3}\varphi$, with φ from K). One groundwater level enters the soil weight, the water pressure and the buoyancy of the body.
3. **Same draws.** Both limit states use the same draws, so their correlation is kept: for type 2, overturning and pile failure occur together 3.4 times as often as if they were independent. The years 2026 and 2041 share the piles; only the shell grows.
4. **Proven strength.** The check "stands today" shares all resistance and soil variables with the future load case, but has its own groundwater and no surcharge.

Other variables are independent, as no data on their correlation exist. We tested the two most likely correlations (Appendix B): denser soil is heavier with a lower K, $\rho(K, \gamma) = -0.5$, and stronger timber is stiffer, $\rho(\mathrm{MOR}, E) = 0.8$ [10]. Each changes β by at most 0.03, mostly upwards, so independence is acceptable and slightly conservative. The walls are simulated independently, although in reality they share the uncertain pile condition, surcharge and model factors; this matters when walls are combined in Section 3.1.

2.1.4 Model quality and validity

Few inputs can be tested against data, but where they can:

- **Werf level.** AHN points on a werf scatter by about 0.08 m, so the wall average is known to about 0.04 m, close to $\sigma = 0.05$ m. For three walls (Rak 03 OZ, Kromme Nieuwe Gracht 002 and 009) the median includes points on stairs or the street, which makes their werf 0.3–0.5 m too high and their β too low.
- **Groundwater.** The BRO wells, interpolated to the walls, give +0.9 to +2.1 m NAP, mostly above the werf itself: they stand under the raised streets and do not describe the werf, which drains to the canal. The range is therefore a physical bound, not a fit. Raising it to $U(+0.6, +1.0)$ m NAP lowers β by 0.04–0.09 (type 2) and 0.1–0.4 (type 1).
- **Soil class.** Only 25–67 % of the borings within 150 m give the class of the nearest boring. Treating the class as uncertain, with weights from these borings, changes β by -0.7 to $+0.3$. Walls classed as sand lose most, as half of their nearby borings find clay or peat. This is the weakest part of the input model.
- **Pile shell.** $U(0, 45)$ mm has a mean of 22.5 mm, close to the 21 mm measured in Amsterdam [7]; no Utrecht data exist.

The strongest test is that all 17 walls stand, so a valid model must let them stand in most draws. A first model that treated the wall as a masonry block failed in almost every draw and was rejected. With the measured sections, type 2 stands in 87–100 % of the draws and type 1 in only 10–98 %: the type-1 model is conservative, or many walls are type 2, and the update shifts $P(T{=}2)$ accordingly. The Monte Carlo error is small (CoV of $\hat{P}_f$ below 2 %, β within ± 0.01).

The model is valid for low walmuren with a werf on their old timber piles, for the two Oudegracht cross-sections. It does not cover settlement, sliding or failure of the timber screen. Because the main inputs are literature values and ranges, not fits to Utrecht data, the results show the ranking and order of magnitude of P_f rather than exact values (Section 4.2).

A Data, data generation and assumptions

This appendix supports Section 1.2. It explains how to rerun the analysis, which walls are modelled, the generated input per wall, every model input, and the data that were considered but not used. All file paths are relative to the project repository.

A.1 Reproducing the results

The scripts below run in this order; each reads the output of the one before. Every raw file in `data/raw/` is listed with its source, URL and download date in `data/raw/SOURCES.md`. Files downloaded from an API (AHN, BGT, NDW, *Werfmuren*) are cached; deleting a cached file makes the script download it again. The Python packages are in `requirements.txt`. The Monte Carlo uses a fixed seed (1) and 400,000 draws per wall and wall type; a full run takes about two minutes.

Table 4: Run order of the scripts.

Step	Script (in `scripts/`)	Output (in `data/processed/`)
1	`build_groundwater_layer.py`	`objects_groundwater.csv`: GHG and GLG per wall
2	`build_soil_type.py`	`masonry_walls_soil.csv`: soil class per wall
3	`build_walmuur_height.py`	`walmuur_height.csv`: werf level and H
4	`build_walmuur_traffic.py`	`walmuur_traffic.csv`: road distance and axle limit
5	`build_walmuur_foundation.py`	`walmuur_foundation.csv`: renewed or timber, year, method
6	`build_masonry_data_gaps.py`	`masonry_walls_data_gaps.csv`: missing fields
7	`table_wervengebied_data.py`	`figures/wervengebied_data_table.md`: data table
8	`StressCaclulation/FailureProbability.py`	`walmuur_failure_probability.csv`, `_draws.json`, `_inputs.csv`

A.2 Which walls are modelled

The register holds 51 named walmuren (4,062 m). Its own design life is 80 years for each of them. A wall past its design life (Aanlegjaar + 80 $<$ 2026) and made of masonry needs a failure probability: 29 walls (1,885 m). The register year is often not updated after a foundation renewal, so we selected on the foundation instead (Table 5).

Table 5: Selection of the walls for the P_f calculation.

Step	Walls	Of which past design life
Named walmuren in the register	51	29
Renewed on steel sheet or tube piles (not modelled)	−32	−11
On timber, but no werf: the street runs to the water (not modelled)	−2	−2
Modelled: on timber piles (or status unclear) with a werf, 611 m	17	16

The two walls without a werf are Kromme Nieuwe Gracht 001 and Oude Gracht Rak 07 OZ; they hold up the street and would need another model. Kromme Nieuwe Gracht 005 is within its design life in the register, but its foundation status is unclear, so it is modelled.

A.3 Generated input per wall

Table 6 gives the values that differ between the 17 walls. All other inputs are the same for every wall (Table 7).

Table 6: Generated input per modelled wall. OG = Oude Gracht, KNG = Kromme Nieuwe Gracht. Werf level from AHN, from the lowest AHN point after a map check (m), or assumed $U(1.0, 1.6)$ m NAP where a werf was seen but no level could be read (a). Status: not renewed, in progress or unclear.

Wall	Number	Length [m]	Werf [m NAP]	Soil class	Status	P(type 2)
OG Rak 03 OZ	KM060052	101.1	1.48	clay/peat	unclear	0.2
OG Rak 04 OZ	KM060054	67.4	1.22	sand	not renewed	0.2
OG Rak 05 NZ	KM060057	72.1	1.22	sand	not renewed	0.2
OG Rak 06 NZ	KM060059	43.7	0.99 (m)	sand	in progress	0.2
OG Rak 06 ZZ	KM060060	57.1	1.21 (m)	sand	unclear	0.2
OG Rak 08 OZ	KM060062	32.7	1.04 (m)	mixed	not renewed	0.8
OG Rak 09 ZOZ	KM060065	54.5	1.23	clay/peat	unclear	0.8
OG Rak 09 WZ	KM060064	53.4	1.28	clay/peat	not renewed	0.8
OG Rak 09 WZ 01	KM060067	3.4	1.13	clay/peat	not renewed	0.8
KNG 002	KM060020	16.2	1.57	clay/peat	not renewed	0.9
KNG 003	KM060021	9.9	1.16	mixed	not renewed	0.9
KNG 004	KM060022	11.1	1.31	mixed	not renewed	0.9
KNG 005	KM060023	13.3	(a)	clay/peat	unclear	0.5
KNG 006	KM060024	4.1	(a)	clay/peat	unclear	0.5
KNG 007	KM060025	10.3	(a)	clay/peat	unclear	0.5
KNG 008	KM060026	42.1	0.99 (m)	clay/peat	unclear	0.5
KNG 009	KM060027	18.3	1.53	sand	unclear	0.5

A.4 All model inputs

Table 7 lists every input of the failure model with its value or distribution and its source. Table 8 gives the soil parameters per class. The model also writes this list to the file `walmuur_failure_inputs.csv` in `data/processed/`.

A.5 How the per-wall data are made

Werf level and H. The AHN ground level is read at six places along each wall, 2.5 m to either side. AHN has no value over water, so the side with values is the werf. The werf level is the median there. Where the ground was above +1.6 m NAP (possibly no werf), the wall was checked by hand on a map and Street View; the lowest AHN point below +1.6 m NAP is then used. The canal bed is not measured anywhere, so H (werf level minus the wall underside at −0.08 m NAP) is a lower bound. The median of 1.13 m NAP on the Oudegracht matches the +0.95 to +1.15 m NAP in [4, 3].

Soil class. Of the 76 DINO borings, the 71 with a known ground level are used. For each wall the nearest boring within 250 m that covers +1.10 to −0.08 m NAP (werf to wall underside, [3]) is taken. Clay-like soil is klei, veen, leem and mergel; sand-like is zand, grind and schelpen; fill and undescribed layers count as neither. The boring is 20–144 m from the wall (median 75 m). For only 9 of the 51 walls do all borings within 150 m give the same class.

Foundation status. The register has no foundation data. We compiled the renewal status, method and year per rak and bank from progress reports, [1], [6], the archaeology reports,

Table 7: Every input of the failure model. Inputs marked * differ per wall (Table 6). CoV = coefficient of variation.

Input	Value or distribution	Source
Geometry		
Werf level*	$N(\text{AHN median}, 0.05\,\text{m})$	AHN [14]
Werf level, no level read*	$U(1.0, 1.6)$ m NAP	Range of the measured werven
Cross-section, type 1 and 2	Measured polygons	[9] Afb. 2.1.1a; [4] Afb. 3.1
Foundation level	Drawing $+U(-0.15, 0.15)$ m	Guess; wall underside -0.08 m NAP [3]
Model factor on M_R	$LN(1, \text{CoV } 0.15)$	Guess (schematic drawings)
Wall type*	$P(\text{type 2})$ per rak	[15] Bijlage G; [9, 4]
Materials		
Masonry	$N(19\,\text{kN/m}^3, 0.05)$	Typical value
Tamped concrete	$N(23\,\text{kN/m}^3, 0.05)$	Typical value
Reinforced concrete	$25\,\text{kN/m}^3$	Typical value
Timber floor	$10\,\text{kN/m}^3$	Waterlogged pine
Soil, water and loads		
Soil class*	K, γ per class (Table 8)	DINO [16]; group report Tab. C.5
Wall friction	$\delta = \frac{2}{3}\varphi$, φ from K	Rankine; common practice
Canal level	$+0.6$ m NAP	[3]
Groundwater under the werf	$U(+0.5, +0.9)$ m NAP	Guess around the canal level
Surcharge q	$N(7\,\text{kPa}, 0.20)$	[8]
Distance a of the surcharge	$U(0, 3)$ m	Guess
Timber piles		
Diameter D_0	$N(180\,\text{mm}, 0.10)$	[9] p. 4; CoV [7] Tab. 7.1
Spacing along the wall	$U(0.8, 1.2)$ m	[9] p. 4, "ca. 1 meter"
Soft shell today	$U(0, 45)$ mm	[7] ch. 3: mean 21 mm, max. 45 mm
Shell growth after 2026	$U(0.10, 0.25)$ mm/yr	[7] §7.3.2: 0.17 mm/yr
Bending strength MOR	$N(22\,\text{MPa}, 0.10)$	[7] Tab. 7.1
Modulus of elasticity E	$N(9\,\text{GPa}, 0.20)$	[7] Tab. 7.1
Subgrade modulus k	$U(4{,}300, 10{,}000)\,\text{kN/m}^3$	[7] Tab. 7.1, clay and peat
Model factor on pile moment	$LN(1, \text{CoV } 0.10)$	JCSS, as [7]
Method		
Proven strength	Wall stands today, no surcharge	[8] eq. 12
Years	2026 and 2041	15-year reference period [13]
Target β	1.8, 2.5, 3.3 (CC1, CC2, CC3)	[13] Tab. B.2, rejection level

Table 8: Soil parameters per class. K is lognormal, γ normal; they are drawn independently. The soil load scales with $K\gamma$.

Soil class	K mean (CoV)	γ [kN/m^3] mean (CoV)	$K\gamma$	Walls (of 51)
Soft clay / peat	0.60 (0.20)	15.0 (0.05)	9.0	23
Mixed	0.45 (0.25)	16.5 (0.06)	7.4	12
Sand	0.35 (0.20)	18.0 (0.05)	6.3	16

city news and neighbourhood letters (`data/raw/walmuur_foundation_sources.csv`, every row with its sources). The city's *Werfmuren* layer links each register wall to its rak and bank. The renewed walls add up to about 3.36 km when the parts still to do are left out, against 3,356 m reported by the city.

Wall type. Two types of low walmuur stand on timber piles [9]. Type 1 (late 19th, early 20th century) is stepped masonry on a timber floor with a timber beam (*watersloof*) on two vertical pile rows. Type 2 (after 1945 until the 1970s) is a brick facing with tamped concrete on a 1.37 m reinforced concrete floor, with the front pile row raked 10:1. The 2008 dive notes [15] record a watersloof in rak 3–6 and a concrete floor in rak 22; rak 21 had both. Rak 7–9 have no note; there, 0.8 follows from [4], which calls the usual 1960s walls largely concrete. TNO does not say which bank a segment is on, so these are chances.

Groundwater and traffic. GHG and GLG per wall are a $1/d^2$-weighted mean of the six nearest BRO wells with a phreatic filter measured after 2020. They are not used for P_f, because the nearest well is 143–418 m away; a range around the canal level is used instead. The road (BGT) lies 2.4–18 m behind the wall, over the werf cellars, and every nearby NDW sign limits axles to 2 t. The load on the werf itself is pedestrians and terraces, so $q \sim N(7\,\text{kPa}, 0.20)$ is on the safe side.

A.6 Data considered but not used

- **BRO CPTs and hand borings:** the CPTs were mostly pre-drilled through the top 2 m, and the hand borings are one site about 650 m from the canals.
- **Traffic counts** in the project folder are from streets outside the Wervengebied.
- **TNO Pilodyn classes** [15] were measured on the screen planks and the watersloof, not on the piles. The pile values (Bijlagen A and B) are blank in the public copy.
- **Piles S13–S19** in [4] belong to an earlier stair, not to the walmuur.
- **A masonry-block resistance** ($\gamma_m H t^2/2$, with t the wall thickness) gave $P_f \approx 1$ for walls that stand, so the measured cross-sections are used instead.

A.7 Limitations of the model inputs

- The two cross-sections are schematic and come from the Oudegracht. The Kromme Nieuwegracht walls (concrete sheet piles in rak 21 and 22) may differ.
- The pile condition is the same broad range for every wall. Measured soft shells per rak would make P_f differ between walls for the right reason.
- K and γ are independent, $\gamma_{\text{sat}} = \gamma$, and the water pressure is hydrostatic. Only one load combination is used.
- Not modelled: sliding, axial pile capacity and settlement, and failure of the timber screen. In 2007–08 divers found the screen in very poor state or gone [9]; without it the soil below the floor also loads the piles.
- Kromme Nieuwe Gracht 005–007 get identical results, because nothing in the data tells them apart.
- The proven-strength evidence is “the wall stands today without surcharge”, not a known load that the wall has survived.

B Probabilistic model

This appendix supports Section 2.1. It gives the equations of the two limit states, the estimation of the parameters, and the checks of the model against data and against changes in the dependence assumptions. The model is `scripts/StressCaclulation/FailureProbability.py`.

B.1 Load on the wall

All moments are taken about the front pile row at the underside of the floor, z_b (-0.55 m NAP for type 1, -0.22 m NAP for type 2, plus Δz). The loaded height is $H = z_{\text{werf}} - z_b$. The groundwater and the canal stand h_w and h_c above z_b (between 0 and H). The load moment is

$$S = \underbrace{\tfrac{1}{6}K\left(\gamma H^3 - \gamma_w h_w^3\right)}_{\text{soil, effective weight}} + \underbrace{\tfrac{1}{6}\gamma_w\left(h_w^3 - h_c^3\right)}_{\text{groundwater}-\text{canal}} + \underbrace{\tfrac{1}{2}Kq_{\text{eff}}H^2}_{\text{surcharge}}, \qquad q_{\text{eff}} = q\,\frac{H^2}{H^2 + a^2}, \tag{3}$$

and the horizontal force is $F_h = \frac{1}{2}K(\gamma H^2 - \gamma_w h_w^2) + Kq_{\text{eff}}H + \frac{1}{2}\gamma_w(h_w^2 - h_c^2)$. The same K and the same groundwater level enter every term.

B.2 Resistance against overturning

The body that tips is the wall, its floor and the soil on the floor. Its weight V and moment M about the front pile row follow from the areas of the parts in the measured cross-section (masonry, tamped concrete, reinforced concrete or timber, and soil). Below the mean of the canal and the groundwater level each part weighs $\gamma - \gamma_w$. The soil behind the body drags it down by wall friction:

$$M_R = \theta_R M + \tan\delta\, F_s\,(x_{\text{back}} - x_{\text{front}}), \qquad \delta = \tfrac{2}{3}\varphi, \quad \sin\varphi = \frac{1-K}{1+K}, \tag{4}$$

with F_s the horizontal force of soil and surcharge. The wall overturns when $M_R < S$.

B.3 Forces in the piles

The rear row carries $R = (M_R - S)/(x_{\text{rear}} - x_{\text{front}})$; the front row carries the rest of the weight, $F = V + \tan\delta\, F_s - R$. For type 2 the front row is raked 1:10 and takes $0.1F$ horizontally. The rest of the horizontal force is shared by both rows, so per pile (spacing s)

$$P = \tfrac{1}{2}(F_h - 0.1F)\,s, \qquad N = Fs\sqrt{1 + 0.1^2}. \tag{5}$$

The moment in a long pile on elastic springs (Hetényi) is

$$M = \frac{P}{2\lambda}\ \text{(fixed head, type 2)}, \qquad M = \frac{0.3224\,P}{\lambda}\ \text{(pinned head, type 1)}, \tag{6}$$

with $\lambda = (kD_0/4EI)^{1/4}$ and $I = \pi d^4/64$. A sound core thinner than 20 mm counts as broken. The pile breaks when

$$\theta_M \frac{32M}{\pi d^3} + \frac{4N}{\pi d^2} > \text{MOR}, \qquad d = D_0 - 2(s_0 + r\,(t - 2026)), \tag{7}$$

where s_0 is today's soft shell and r its growth rate.

B.4 Parameter estimation

Moments to distribution. A lognormal with mean m and CoV v has $\sigma_{\ln}^2 = \ln(1 + v^2)$ and $\mu_{\ln} = \ln m - \sigma_{\ln}^2/2$. Normal and uniform distributions are set by their mean and CoV, or by their bounds.

Proven strength. Let E be the event that the wall stands today under soil and water, without surcharge. Every draw $\mathbf{x}_i$ is checked against E with its own resistance and soil and a separate groundwater draw. Bayes' rule gives $f(\mathbf{x} \mid E) \propto f(\mathbf{x})\,\mathbf{1}[\mathbf{x} \in E]$, so the updated failure probability is estimated by rejection:

$$\hat{P}_{f|E} = \frac{\sum_i \mathbf{1}[\text{fail}_i]\,\mathbf{1}[E_i]}{\sum_i \mathbf{1}[E_i]}. \tag{8}$$

Wall type. With prior $p = P(T = 2)$ per rak, the same evidence updates the type, and the best estimate weighs the two types:

$$P(T = 2 \mid E) = \frac{p\,P(E \mid T = 2)}{p\,P(E \mid T = 2) + (1 - p)\,P(E \mid T = 1)}, \qquad P_f = \sum_T P(T \mid E)\,P_f(T \mid E). \tag{9}$$

B.5 Sensitivity to dependence and inputs

The script `SensitivityDependence.py` (in `scripts/StressCaclulation/`) reruns the model with one change at a time, with 200,000 draws per wall, type and variant and the same seeds. It writes `walmuur_sensitivity.csv` in `data/processed/`. The correlations use a Gaussian copula: the two variables are drawn from correlated standard normals and transformed to their own distributions. For the uncertain soil class, every class found within 150 m gets a prior weight equal to its share of the borings; like the wall type, it is then updated with proven strength. Table 9 summarises the change in β (updated, 2026); Table 10 gives every wall.

Table 9: Change in β against the base model: median [range] over the 17 walls. Best estimate: the two wall types weighted by $P(T \mid E)$.

Change	Best estimate	Type 1	Type 2
$\rho(K, \gamma) = -0.5$	0.00 [−0.01, +0.01]	+0.01 [−0.01, +0.02]	0.00 [−0.01, +0.01]
$\rho(\text{MOR}, E) = +0.8$	0.00 [−0.01, +0.01]	0.00 [−0.03, +0.01]	+0.01 [0.00, +0.03]
Groundwater +0.6 to +1.0 m NAP	−0.02 [−0.38, +0.14]	−0.20 [−0.43, −0.12]	−0.06 [−0.09, −0.04]
Soil class uncertain	+0.06 [−0.70, +0.29]	–	–

Correlations. Both correlations change β by less than the Monte Carlo error of these runs (± 0.01) for most walls, and by at most 0.03. A negative $\rho(K, \gamma)$ narrows the spread of $K\gamma$, which drives the soil load. A positive $\rho(\text{MOR}, E)$ couples strength to stiffness: a stiffer pile attracts a larger moment ($M \propto E^{1/4}$), but its higher MOR more than covers this.

Groundwater. Higher groundwater adds water pressure behind the wall, so β falls for each type. For the best estimate the effect is partly offset: under the higher water fewer type-1 draws stand today, so the update moves $P(T{=}2)$ up. Walls in sand, where type 1 is likely, lose most.

Soil class. Most walls have borings of all three classes within 150 m. Walls classed as sand lose up to 0.7 in β, because half of their nearby borings find clay or peat. Walls classed as clay or peat gain up to 0.3.

The two limit states. For type 2, overturning and pile failure occur together 3.4 times as often (median) as if they were independent: both grow with the soil load. Because overturning is rare for type 2, the series P_f differs by at most 8 % from the value for independent limit states.

Table 10: β per wall (updated, 2026, 200,000 draws) for each change. Shares c/s/m: clay/peat, sand and mixed among the borings within 150 m. OG = Oude Gracht, KNG = Kromme Nieuwe Gracht.

Wall	Class	Shares c/s/m	Base	$\rho(K,\gamma)$	$\rho(\mathrm{MOR},E)$	Groundw.	Soil unc.
OG Rak 03 OZ	clay/peat	0.50 / 0.50 / 0.00	0.13	0.14	0.12	0.27	0.34
OG Rak 04 OZ	sand	0.50 / 0.33 / 0.17	0.81	0.81	0.81	0.55	0.38
OG Rak 05 NZ	sand	0.40 / 0.60 / 0.00	0.81	0.81	0.81	0.55	0.55
OG Rak 06 NZ	sand	0.50 / 0.33 / 0.17	1.18	1.17	1.18	0.80	0.48
OG Rak 06 ZZ	sand	0.57 / 0.29 / 0.14	0.82	0.82	0.82	0.57	0.33
OG Rak 08 OZ	mixed	0.56 / 0.11 / 0.33	1.38	1.38	1.38	1.30	1.28
OG Rak 09 ZOZ	clay/peat	0.50 / 0.17 / 0.33	1.08	1.08	1.09	1.12	1.19
OG Rak 09 WZ	clay/peat	0.56 / 0.11 / 0.33	1.05	1.05	1.05	1.08	1.14
OG Rak 09 WZ 01	clay/peat	0.50 / 0.17 / 0.33	1.13	1.13	1.14	1.19	1.26
KNG 002	clay/peat	0.67 / 0.00 / 0.33	0.79	0.78	0.80	0.75	0.90
KNG 003	mixed	0.75 / 0.00 / 0.25	1.52	1.52	1.53	1.48	1.37
KNG 004	mixed	0.75 / 0.00 / 0.25	1.38	1.38	1.39	1.36	1.22
KNG 005	clay/peat	0.50 / 0.00 / 0.50	0.59	0.59	0.59	0.67	0.65
KNG 006	clay/peat	0.40 / 0.20 / 0.40	0.59	0.60	0.60	0.67	0.73
KNG 007	clay/peat	0.40 / 0.20 / 0.40	0.59	0.59	0.59	0.67	0.73
KNG 008	clay/peat	0.40 / 0.20 / 0.40	0.56	0.56	0.56	0.60	0.85
KNG 009	sand	0.67 / 0.33 / 0.00	0.72	0.72	0.73	0.66	0.66

B.6 Checks against data

Table 11 compares the input distributions that can be checked with the data available for the 17 walls.

Table 11: Input distributions against the available data, for the 17 walls.

Input	Model	Data	Verdict
Werf level	$N(\text{median}, 0.05\,\text{m})$	AHN points below +1.6 m NAP: SD 0.08 m (0.01–0.33) per wall; standard error of the mean about 0.04 m	Fits the wall average; three walls biased by stair or street points
Groundwater	$U(+0.5, +0.9)$ m NAP	BRO, $1/d^2$ over 6 wells, nearest 146–309 m away: GLG +0.90 to +1.64, GHG +1.43 to +2.15 m NAP	Data not representative of the werf; physical bound used
Soil class	Class of nearest boring	25–67 % (median 40 %) of borings within 150 m agree	Weak; class should be uncertain
Soft shell	$U(0, 45)$ mm, mean 22.5	Amsterdam pine piles: mean 21 mm, max. 45 mm [7]	Fits the measured range; no Utrecht data
Surcharge	$N(7\,\text{kPa}, 0.20)$	No load data; 2 t axle limit and a pedestrian werf	Not testable; on the safe side

B.7 Monte Carlo accuracy

With n draws that stand today, the coefficient of variation of the updated estimate is

$$\mathrm{CoV}(\hat{P}_f) = \sqrt{\frac{1-\hat{P}_f}{n\hat{P}_f}}. \tag{10}$$

The worst case is Rak 06 NZ, type 2, with $\hat{P}_f = 0.007$ and $n = 398{,}840$: a CoV of 1.9 %. The standard error of β is at most 0.01 for every wall and type.

References

[1] Antea Group. Onderzoek wal- en kluismuren, eindrapportage. Technical report, Antea Group, November 2019.

[2] Basisregistratie Ondergrond. BRO groundwater monitoring wells and head series. PDOK OGC API, `https://api.pdok.nl/tno/bro-grondwatermonitoring-in-samenhang-karakteristieken/ogc/v1`. Downloaded 16 September 2026.

[3] M. Duurland. Oudegracht rakken 1 oost, 3 west en 12 oost. Basisrapportage Archeologie 158, Erfgoed gemeente Utrecht, 2019.

[4] M. Duurland. Oudegracht rakken 1, 4, 10, 11 en 15 west. Basisrapportage Archeologie 156, Erfgoed gemeente Utrecht, 2020.

[5] Gemeente Utrecht. Output areaal GU TU Delft (asset register of quay walls and related objects), 2025. GeoJSON, 3,817 objects, delivered to TU Delft for the CME4150 team challenge on 26 August 2025.

[6] Gemeente Utrecht. Programma werk aan de werf, uitvoeringsprogramma 2025–2028. Technical report, Gemeente Utrecht, 2025.

[7] M.-J. Hemel. *Amsterdam quays under pressure: modelling and testing of historic canal walls*. PhD thesis, Delft University of Technology, 2023.

[8] M.-J. Hemel, D. J. Peters, T. Schweckendiek, and S. N. Jonkman. Reliability updating for lateral failure of historic quay walls. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, 18(4):882–903, 2024.

[9] IBU Stadsingenieurs. Ontwerpnotitie herstel lage walmuren, restauratie wal- en kluismuren. Technical report, Gemeente Utrecht, c. 2010.

[10] Joint Committee on Structural Safety. JCSS probabilistic model code. Technical report, JCSS, 2001. Part 3.5: Properties of timber; Part 3.9: Model uncertainties.

[11] Kadaster. Basisregistratie grootschalige topografie (BGT), road areas (wegdeel). PDOK OGC API, `https://api.pdok.nl/kadaster/bgt/ogc/v1`. Downloaded 21 October 2026.

[12] Nationaal Dataportaal Wegverkeer. Traffic signs, current state. `https://data.ndw.nu/api/rest/static-road-data/traffic-signs/v5/current-state`. Downloaded 21 October 2026.

[13] NEN. NEN 8700:2011+A1:2020: Beoordeling van de constructieve veiligheid van een bestaand bouwwerk bij verbouw en afkeuren – grondslagen, 2020.

[14] Rijkswaterstaat. Actueel hoogtebestand nederland (AHN), digital terrain model 0.5 m (dtm_05m). PDOK WMS, `https://service.pdok.nl/rws/ahn/wms/v1_0`. Accessed 20 October 2026.

[15] TNO. Eindbeoordeling kademuren Utrecht. Technical Report TNO-034-DTM-2009-01978, TNO Bouw en Ondergrond, May 2009.

[16] TNO Geologische Dienst Nederland. DINO geological borehole logs, Utrecht city centre. BROloket, `https://www.broloket.nl`. Downloaded 14 October 2026.